

Python Practice Handout

Lists

1. What is a list

A list is a new type. It holds several pieces of information in a set order. Favourite foods, the results of a race, or a to-do list are all good examples. The order is the important part.

You make a list with square brackets. Empty brackets make an empty list.

```
fav_movies = []
```

2. Putting items in a list

Type the items inside the brackets and separate them with commas. A list can hold ints, floats, Booleans or strings.

```
fav_numbers = [1, 5, 7, 3]
mixed = [1, 5.5, True, "hello"]
```

Movie titles are text, so they go in quotes.

```
fav_movies = ["Sandlot", "The Lego Movie", "Dune"]
```

3. Printing a list

Printing the whole list shows it much as you typed it.

```
fav_movies = ["Sandlot", "The Lego Movie", "Dune"]
print(fav_movies)
```

Output:

```
['Sandlot', 'The Lego Movie', 'Dune']
```

4. Getting one item out

Put square brackets after the list name and say which item you want. This number is called the index. Counting starts at zero, not one.

So Sandlot is item 0, The Lego Movie is item 1, and Dune is item 2.

```
fav_movies = ["Sandlot", "The Lego Movie", "Dune"]

print(fav_movies[0])
print(fav_movies[1])
print(fav_movies[2])
```

Output:

```
Sandlot
The Lego Movie
Dune
```

This catches everyone at first. Asking for item 1 gives you the second thing in the list.

5. Try it yourself

Make a list of your three favourite numbers and print the first one. Try this on your own before looking below.

```
fav_numbers = [4, 11, 27]

print(fav_numbers[0])
```

Output:

```
4
```

6. How long is a list

Use len, short for length, with the list name inside the brackets. Print it to see the answer.

```
fav_movies = ["Sandlot", "The Lego Movie", "Dune"]

print(len(fav_movies))
```

Output:

```
3
```

7. Adding to the end

Write the list name, a dot, then append. Whatever you put in the brackets is added on the end of the list.

```
fav_movies = ["Sandlot", "The Lego Movie", "Dune"]

fav_movies.append("Iron Man")

print(fav_movies)
print(len(fav_movies))
```

Output:

```
['Sandlot', 'The Lego Movie', 'Dune', 'Iron Man']
4
```

8. Adding at a chosen place

Use insert when the new item should not go on the end. Give it two things: the index where it should go, then the item itself.

An index of 1 puts it in second place, just after Sandlot.

```
fav_movies = ["Sandlot", "The Lego Movie", "Dune", "Iron Man"]

fav_movies.insert(1, "Batman")

print(fav_movies)
```

Output:

```
['Sandlot', 'Batman', 'The Lego Movie', 'Dune', 'Iron Man']
```

9. Removing an item

Write `del`, then the list name with the index of the item you want gone. The Lego Movie is now at index 2.

```
fav_movies = ["Sandlot", "The Lego Movie", "Dune"]
fav_movies.append("Iron Man")
fav_movies.insert(1, "Batman")

del fav_movies[2]

print(fav_movies)
```

Output:

```
['Sandlot', 'Batman', 'Dune', 'Iron Man']
```

Every item after the deleted one shifts down, so the indexes change. Print the list after each change to see where things are.

10. Try it yourself

Remove items from the list until only one movie is left. The simplest way is to delete item 0 over and over. Try this on your own before looking below.

```
fav_movies = ["Sandlot", "Batman", "Dune", "Iron Man"]

del fav_movies[0]
del fav_movies[0]
del fav_movies[0]

print(fav_movies)
```

Output:

```
['Iron Man']
```